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The Brazilian biofuel policy (RenovaBio) and its uncertainties: An assessment of technical, socioeconomic and institutional aspects

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ABSTRACT

The National Biofuels Policy (RenovaBio) was launched in December 2017 with the aim of supporting the Brazilian Nationally Determined Contribution (NDC)'s commitments under the Paris Agreement. This policy includes mechanisms for Life Cycle Assessment (LCA), commercialization and predictability of the fuel market, supporting national energy security and reducing greenhouse gas (GHG) emissions. It focuses on increasing the biofuel consumption and its expansion in the Brazilian energy matrix, giving incentives to producers, and creating an open market of carbon-reduction credits called CBIO. In this context, this paper aims to assess the RenovaBio program and its main challenges from a technical, economic and institutional perspective, besides assessing the lessons learned from the prior national biofuel policies. We analysed the scientific literature from Scopus and ScienceDirect databases by using the T-Lab's topic model tool, which combines the Latent Dirichlet Allocation (LDA) algorithm and Gibbs sampling, so as to provide a way of identifying, examining, and modelling topics from the texts. Our conclusions put light on the need to establish RenovaBio's legal and regulatory framework, in addition to reassessing the LCA methodology and eligibility details, as concerns arose over land use displacement for sugarcane and traceability of agricultural feedstock. Furthermore, the Program fostered discussions on new technologies and the regulated carbon market in the fuel sector in Brazil, in addition to the social aspect, especially considering the pandemic period of the program implementation and its impacts on the fluctuation of agribusiness commodities prices and fuel sales.

Introduction

The amount of carbon dioxide in the atmosphere is mostly caused by fossil fuels burned for electricity, heat and transport, which are contributing to climate change, especially in developing countries [1,2]. Additionally, in the last few years, fossil fuels are facing criticism due to the high utilization by the dense population, urbanization of various geographical areas and industrialization which directly affect the global climate and economy. Thus, with the expansion of fossil fuels and their hazards, fuels from biological origin such as ethanol, vegetable oil and various biomasses are gaining attention. In this regard, biofuels environmental benefits are widely touted among renewable fuels' advocates to reduce pollution and climate-altering greenhouse gas (GHG) emissions worldwide [3,4].

Sorda et al. (2010) review governmental policies and national strategic plans for biofuels in different countries, and their expansion is basically justified by government interventions such as financial incentives, blending mandates and feedstock use, sustained by environmental strategies [5]. These strategies combined with intensive development of biofuels are based on the assumptions, for example, that the production of ethanol from sugarcane and sugar beet could cut emissions between 60% and 90%. Meanwhile, production of biodiesel, maize and palm oil, and rapeseed turn out to reduce emission by 35%, 80% and 60%, respectively [1].

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Particularly in Brazil, programs related to the production of biofuels, in particular ethanol from sugarcane, has gone through various phases, since 1975 when the country's military regime initially launched the National Alcohol Program (Proálcool) to meet geopolitical and energy security objectives. Ethanol production from sugarcane was spurred by a need to enhance the trade balance and national energy self-sufficiency in the context of an economically deleterious trade deficit provoked by dependence on foreign oil and the 1970s international oil crisis [3,6]. Furthermore, the National Program of Production and Use of Biodiesel (PNPB) launched in 2005 incorporates footsteps of Proálcool focusing on biodiesel supply, also with a strong focus on social aspects and regional development [7]. Notably, both programs were created emphasizing supply and energy security as primary goals, while biofuels environmental benefits were recognized later [3,8].

In 2019, the transport sector in Brazil responded to approximately 33% of the total energy consumption and it was responsible for 190,5 $MtCO_2eq$ – 45,4% of total emissions from the energy sector [9]. Also, transport has oil as its primary source of energy, a large emitter of polluting gases [10]. In this sense, the National Biofuels Policy (RenovaBio), created in December 2017, introduces a methodology that allocate costs on externalities to achieve environmental targets, as well as gives incentives for the expansion of biofuel production and its use for transportation [3,11,12].

In brief, RenovaBio was launched aiming at meeting Brazil's commitments under the Paris Agreement. This policy includes mechanisms for Life Cycle Assessment (LCA), commercialization and predictability of the fuel market, in addition to increasing the use of biofuels, supporting national energy security, and reducing GHG emissions. The main purpose is to increase biofuel consumption and its expansion in the Brazilian energy matrix based on a solid decarbonizing policy, giving incentives to producers, and creating an open market of carbon-reduction credits called biofuel Decarbonization Credit (CBIO) [11,12].

However, Covid-19 dissemination is taking part in the global backdrop and its final impacts are still unknown. Combining the drop of fuel demand during the first periods of isolation, agribusiness sector in Brazil faced initially ethanol and sugar's low prices, contracts cancellation between producers and distributors, changes in the production mix, as well as operational limitations in plants, ports, distribution bases, and financial problems, especially in terms of production costs [10]. For the year 2021, Brazil is still facing severe volatility in the international oil and gasoline prices, and, combined with a rise of commodities exportation and lack of precipitation which led to one of the worst droughts in the country affecting crop and livestock production, brought direct impacts on domestic price of fuels to Brazilian consumers. From the socioeconomic point of view, these impacts transcend the fuel market, leading to higher food and electricity prices, deprivation of basic products and inflation. As observed, the government should propose mechanisms to minimize these cost impacts and negative externalities focusing on the system efficiency [13,14].

In the face of so many uncertainties, market fluctuations and socioeconomic scenario, there is a trigger to review the policies associated with biofuels in Brazil - such as RenovaBio, Biodiesel Auction, Resolutions from the Brazilian Petroleum, Natural Gas and Biofuels Agency (ANP), taxation, financing, among others. Despite the challenges imposed to this new public policy in the energy sector, RenovaBio stimulates the increase of biofuels in a country that is a reference in bioenergy and supports the achievement of the Brazilian Nationally Determined Contribution (NDC)'s goals. So far, the program has not yet been implemented with full effectiveness, even with 14.5 million tons of carbon dioxide avoided by the transport sector as a result of 2019/2020 target [15]. However, it is expected that the government addresses some of RenovaBio's gaps in order to guarantee Brazilian climate's ambitions.

Therefore, this paper discusses the macroeconomic scenario to achieve the mandatory RenovaBio's targets and other topics that may impose barriers to its full implementation. Firstly, this paper brings an overview of the Brazilian biofuel policies, emphasizing RenovaBio structure and giving support to policymakers in terms of scenario and perspectives of the new biofuel market in Brazil. Additionally, it can support researchers grasp the current research and implications about RenovaBio and Brazilian energy policies. For that, we analysed the scientific literature from Scopus and ScienceDirect databases by using the T-Lab's topic model tool, that combines the Latent Dirichlet Allocation (LDA) algorithm and Gibbs sampling, so as to provide a way of identifying, examining, and modelling topics from the texts. Such analysis brought the most relevant topics directly associated with RenovaBio and, considering the main purpose of this paper, we have chosen four of them to be deeply assessed, as further detailed in sections 3 and 4.

The paper is organized as follows. After this introduction (Section 1), we present the development of biofuel policies and discuss the unroll of RenovaBio program in the context of energy policies in Brazil in Section 2. Then, Section 3 describes the methodological procedure used in this study. Section 4 presents and discusses the main results from the developed method, evaluating the main uncertainties about the RenovaBio program, in terms of technical, economics and institutional perspectives. Section 5 brings recommendations that could drive the success of the Program and finally, the main conclusions are presented in Section 6.

Brazilian biofuel policies and the RenovaBio program

Considering the biofuel agenda, Brazil has been a pioneer worldwide [14]. Over the past years, Brazil has implemented ambitious biofuel policies, such as Proálcool in the 70 s and the National Program of Production and Use of Biodiesel (PNPB) in 2004, although both programs seem contradictory, as environmental and social inclusion are not seen as a deep result as energy and agricultural benefits do [7]. In other words, Stattman et al (2013) compare governance fields of bioethanol and biodiesel policies in Brazil and highlight that both Proálcool and PNPB directly depended on energy and agricultural sectors and their framework [7].

First, the regulatory elements that initially shaped Brazil's pioneering in ethanol production were held between the 1920 s and 1950 s, before the implementation of the National Fuel Alcohol Program (Proálcool), when a set of technological Institutes and the presence of the national petroleum company (Petrobras) were responsible for infrastructure and enable the State's control on ethanol production capacity [14,16,17]. In 1964, the military regime was installed, which increasingly stimulated decisions towards the reduction of Brazilian's dependency on imports of fossil fuel [18].

In 1975, the National Alcohol Program (Proálcool) was implemented aiming to address national energy security and supply, generating opportunities for sugar-cane industry [14,18]. Still in the 70 s, the Brazilian Government began mechanisms to encourage vehicles powered by alcohol, and in addition, automakers developed gasoline-powered cars, with the addition of a percentage of anhydrous alcohol. At the end of the 80 s, with the supply crisis caused by the combination of sugar's good remuneration in the international market and the disincentive to alcohol production, also with the reduction of subsidies, did hydrated alcohol lose its competitiveness against gasoline. In the 1990 s, the Brazilian automobile industry was exposed to global competition, and from that moment, the Brazilian market turned to the production of flex-fuel vehicles [19]. Apart from that, between 2003 and 2008, the oil prices contributed to the resurgence of ethanol, as it was a cheaper fuel and the flex-fuel technology allow different fuel choices. In resume, the success of the Proálcool program is a result of the importance of sugar and ethanol production in the Brazilian economy, also, the development and stabilization of flex-fuel motors [5].

Based on the experience of Proálcool, PNPB was implemented by the Federal Law 11.097 of 13 January 2005. Differently from ethanol, biodiesel is subsidized, supported by purchase auctions committed to ensure national supply targets, and tax exemptions, given by a

percentage of agricultural feedstock purchased from family farmer [5]. PNPB initially required 2% of diesel to be replaced by biodiesel from 2008 to 2012 and increase to 5% from 2013 onwards [20], reaching 15% by 2023. Also, as soy is one of the principal agricultural feedstock to biodiesel production in Brazil, there is a discussion regarding its traceability [21], and role as a global commodity [7]

Recently, the National Biofuels Policy (RenovaBio) was launched in order to consolidate those previous biofuel policies and solve national struggle in view of environmental targets, also, creating a new biofuel trading market. RenovaBio was developed as a result of the Brazilian Government's ratification of the Paris Agreement in 2016, where the country committed itself to increase its share of biofuels in the Brazilian energy matrix, by establishing a target to promote a reduction in its GHG emissions in 37% below 2005 levels in 2025, in addition to inform a subsequent indicative contribution of 43% reduction below 2005 emission levels in 2030 [12]. It can be summarized that RenovaBio materializes the institutional structure that encourages the increase in the production of biofuels by valuing the environmental externalities generated for efficient production and by using such products, without any impacts or subsidies that burden public finances [12].

The emission reduction targets for the fuel matrix were defined for the period from 2019 to 2029 by CNPE Resolution No. 15, of June 24, 2019 [22]. These targets will be broken down annually into mandatory individual targets for fuel distributors, according to their participation in the fossil fuel market, under the terms of ANP Resolution No. 791/2019, of June 12, 2019 [23]. Through the certification of biofuel production, different grades will be assigned to each biofuel producer and importer, in a value inversely proportional to the carbon intensity of the biofuel produced [12]. The note will accurately reflect the individual contribution of each producing agent to mitigate a specific amount of GHG in relation to its fossil substitute (in terms of tons of CO_2 equivalent - CO_2e).

Biofuel producers and importers who wish to join the program will hire certifying firms accredited by the ANP to carry out the Biofuel Certification and validate the Energy-Environmental Efficiency Rating (NEEA). The Certificate of Efficient Production of Biofuels will be valid for three years, counting from the date of its approval by the ANP, and can only be issued by the inspecting firm after the approval of the process by the ANP [24,25].

To achieve its objectives, RenovaBio is designed to introduce market mechanisms to recognize the capacity of each biofuel to reduce emissions, individually, per production unit. Basically, there are two main instruments [12,22,23]:

1. Establishment of national emission reduction targets for the fuel matrix, defined for a period of 10 years. The goals are important to bring predictability, in terms of the volumetric need for fuels (fossil and renewable) in this time horizon, and thus allow private agents to make their investment planning and analysis in an environment with less uncertainty.
2. Certification of the production of biofuels, assigning different grades for each producer (higher grades will be given for the producer that produces the greatest amount of net energy, with the lowest CO_2 emissions, in the life cycle). The note accurately reflects the individual contribution of each producing agent to mitigate a specific amount of GHG in relation to its fossil substitute (in terms of tons of CO_2e).

The connection of these two instruments occurs with the creation of the CBIO, which is equivalent to one ton of carbon and it is a fundamental component of the RenovaBio program [12]. This is an asset, traded on a stock exchange, issued by the biofuel producer, from commercialization (invoice). Fuel distributors will meet the target by demonstrating ownership of CBIOs in their portfolio. Its commercialization is an incentive for biofuel producers to expand their production and a way to attract new investments, inducing a greater participation of biofuels in the matrix and contributing to the reduction of GHG emissions.

The annual mandatory target represents the volume of carbon dioxide that must not be emitted to the atmosphere by a renewable fuel or emitted by a fossil fuel, as one ton of carbon dioxide corresponds to one carbon-reduction credit (CBIO). To issue a CBIO, biofuel producers/importers must be certified by an inspection company, which will give them an energy efficiency grade calculated using RenovaCalc, a specific calculator that considers, beyond other aspects, the production route, the way of production, and biomass eligibility criteria [11].

Fig. 1 presents RenovaBio's operating model, with national emission reduction targets being broken down into individual targets for regulated agents (fuel distributors) and the issuance of CBIO by certified biofuel producers/importers. As can be seen, the program has three main instruments: Biofuel production certification, carbon-reduction credits (CBIO) and annual mandatory GHG emission reduction targets.

So far, more than 200 producers have already been admitted to RenovaBio. In order to guarantee the credibility of the CBIO, it is registered by financial institutions and traded on the Stock Exchange. CBIO, in practice, constitutes an element of pricing for carbon reduction. Distributors will have the obligation to purchase CBIOs to meet their goals, that will be issued by producers/importers of biofuels, duly certified by a certifying company, in an amount proportional to the respective volumes sold of these products and to NEEA. Thus, the program is a market-driven scheme with no subsidies and it drives to sustainable bioenergy production [11,12,24-26].

Furthermore, the biofuel producers/importers will have up to 60 days to issue the CBIO after its sale, and the primary issuance of CBIO will be made in book-entry form - contracting a bookkeeper. The negotiation of decarbonisation credits will be carried out in organized markets, and the bookkeeper will be responsible for maintaining the business records that occurred during the period in which the credits were registered. Then, the CBIO will be traded on an exchange by bookkeeper or broker, and can be acquired by distributors, investors, hedge funds and other market players. By the end of the day, the final negotiations will be shown by the stock exchange [10,12].

Considering the annual target for each year, it will be divided into individual targets, and applied to all fuel distributors proportionately to their market share in fossil fuel sales in the previous year. The distributor is obligated to acquire CBIO, otherwise there is a fine that can reach R$ 50 million (approximately US 9,2 million)[4], and the CBIO non-acquired will take part into the target the year forward [12]. It should be noted, however, that, given the current pandemic context of the new coronavirus (Covid-19) and the oscillation of fuel consumption, many questions regarding the volume of CBIOs appear, as well as their estimated price. Such debate will be further developed in Section 5.

Methodological procedure

We began this study by conducting a keyword search on the terms "RenovaBio '' and "biofuel policy Brazil'' in Scopus and ScienceDirect databases to identify articles published on this topic. Our objective is to identify the state-of-the-art of research on RenovaBio and main themes related to this policy that are being addressed by the scientific literature. In Scopus we found a total of 23 articles and 95 articles in ScienceDirect until July 2021. We did not consider other types of documents such as encyclopedias or book chapters, because we did not have access to all of them. After cleaning the duplicate articles, by using the OpenRefine tool, a total of 101 articles remained for the analysis.

Then, we separate the abstracts of these articles into a corpus to proceed with the textual analysis. The corpus was analyzed using T-Lab's topic model tool, that combines a very sophisticated algorithm

[4] Considering the exchange rate of the Central Bank of Brazil on September 29, 2021.

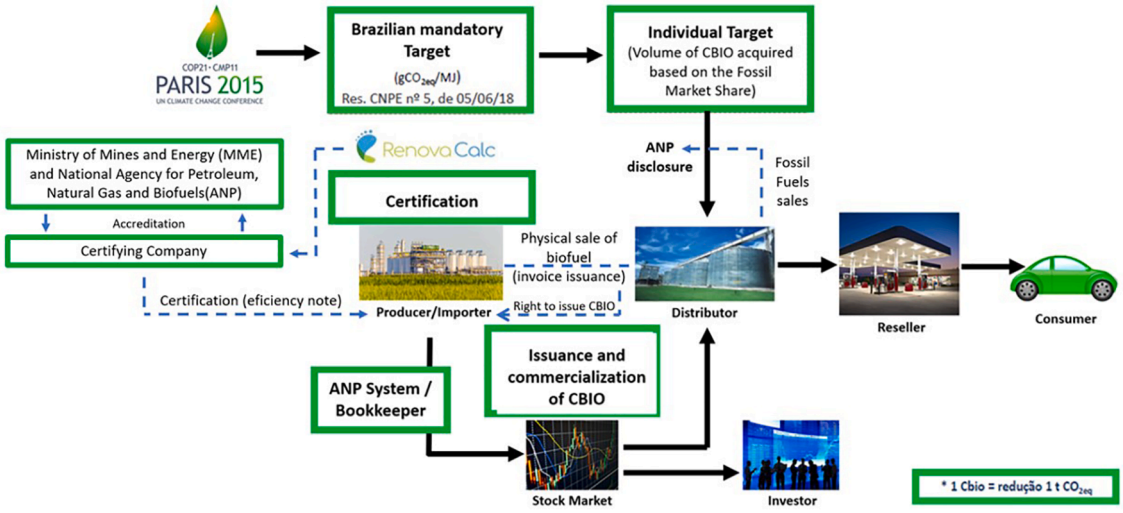


Fig. 1. RenovaBio Flowchart. Source: Own elaboration based in [12].

Latent Dirichlet allocation (LDA) [27] and Gibbs sampling [28], that provides a way of identifying, examining, and modeling topics from the texts [29]. "LDA is a generative probabilistic model of a corpus. The basic idea is that documents are represented as random mixtures over latent topics, where each topic is characterized by a distribution over words" [27].

The topic model enables the description of themes by means of the probability of their characteristic words, either specific or shared by two or more themes. By using T-Lab' topic model tool, 11 emerging topics arrose, so we carried out the following operations: explore the characteristics of each theme; the relationships between the various themes; assess the semantic coherence of each theme; test the model and apply the model by creating a new thematic variable, the values of which are the chosen topics; export a dictionary of categories, which was used for analyses using other tools [30].

Results and discussions

The results are based on the topic-modelling methodology that uses the correspondence analysis to depict the similarity/dissimilarity of themes. This process brings the most relevant topics in the published articles related to a specific analysed topic. As a result, eleven (11) relevant topics emerged from the published articles regarding RenovaBio, as can be seen in Fig. 2.

Fig. 2 shows the eleven (11) themes from the topic model that were aggregated into four (4) different colours: red, blue, turquoise, and fuchsia. The definition of these colors takes into account the relationship between the themes resulting from the methodological process. For example, the theme *Policy* and *Investment* are colored in red. *Policy* emphasizes the creation and establishment of RenovaBio to increase the stability of the biofuels market and to attend the Brazilian Nationally Determined Contribution (NDC), while *Investment* focuses on the relevance of developing reliable policy and decision-making systems in order to stimulate investment in low-carbon technologies. As can be seen, both themes are strongly positively correlated. Topics such as investment for bioenergy generation capacity in sugarcane mills, the feasibility of expanding first-generation sugarcane distillery and second-generation ethanol, and increasing electricity production, besides how research institutions and policymakers can organize for better strategic action, identifying research fields and the need for R&D investment, emerged from these themes.

The themes *Energy* and *Emission* are in blue. *Energy* emphasizes the

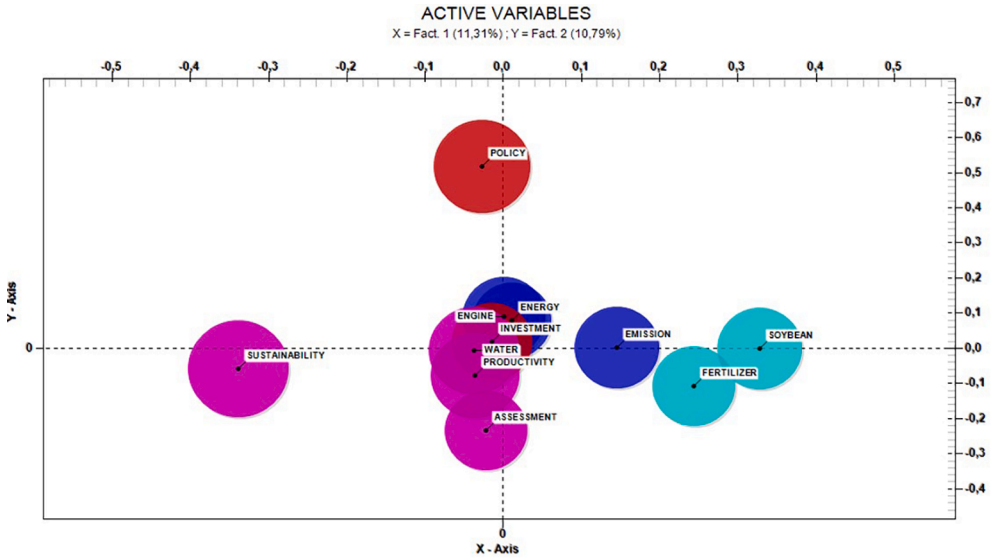


Fig. 2. Results based on the top-modelling methodology. Source: T-Lab' topic model tool.

use of energy from biomass and bioethanol in reducing GHG emissions and the diversification of the energy matrix. Studies show the potential of ethanol and the emissions avoided. *Emission* focuses on the relevance of substituting fossil fuel for biofuel by analysing their carbon intensity and its impacts into grades for facilities, which progressively emit CBIOs and attract investments.

Soybean and *Fertilizer* themes are in turquoise and they depict the use of vinasse as a fertilizer, which is a major byproduct of the sugarcane biofuel industry recycled in the fields. The sugarcane plantation intercropped with soybean and their effects in enzyme activity, nitrogen and phosphorus contents, and the microbe quantity of rhizosphere soil. These two themes are configured as challenges for producers regarding the traceability of the production and, consequently, for LCA, whose origin comes from several suppliers, when not imported. Besides, fertilizers are one of the biggest contributors of GHG emissions, contributing significantly, especially depending on the nitrogen source used.

The themes in fuchsia - *Sustainability* shows the sustainability practices of the biofuels sector. The theme *Engine* is related to flex-fuel engines running on ethanol. The theme *Water* addresses water demand triggered by sugar crops expansion, the irrigation expansion in the rainfed dominant cropping areas that would partially slow down the land-use expansion by boosting yields, and it is also related to productivity. The theme *Assessment* is related to the LCA, which was adopted to support the RenovaBio program that considers the biofuels fuel production life cycle (Well to Wheel).

However, considering the main purpose of this paper, which is to evaluate the technical, socioeconomic, and institutional aspects from RenovaBio, we decided not to analyse all these themes, but instead to carry out a deep analysis of the four following themes: Policy, Assessment, Investment and Emission.

Policy: Targets, socioeconomic scenario, and sales uncertainties

Even before the official announcement of the coronavirus outbreak as a world pandemic by the World Health Organization (WHO), March of 2020 was marked by a price war on oil between Saudi Arabia and Russia. It was triggered by a break-up in dialogue between the Organization of the Petroleum Exporting Countries (OPEC) and Russia over the proposal of oil-production cuts during the COVID-19 pandemic. Thus, rising stockpiles of crude threatened to overwhelm storage facilities, leading prices of Brent crude and US West Texas Intermediate (WTI) sank to levels not seen since 2002, as oil producers were forced to pay buyers to take the barrels they could not store. Meanwhile, in Brazil, this volatility was strongly felt, and the shares of the principal companies, especially those who depended on oil in their operations declined [31].

Prices of biofuels in Brazil have a strong connection to international events, as domestic oil pricing and its derivatives by Petrobras depend on macroeconomic variables, such as commodities future contracts, foreign exchange, international parity, port, and tax bureaucracy. Also, it depends on internal political factors, such as taxes, inflation, and government cash. Therefore, during the first months of Covid-19 pandemic, agribusiness in Brazil faced ethanol and sugar's low prices, contracts cancellation between producers and distributors, changes in the production mix, as well as operational limitations in plants, ports, distribution areas, and financial problems, especially in terms of production costs [10,32,33].

The National Biofuel Policy started to take effect in 2020, but in a scenario of confinement, closure of economic activities, and need to relieve the pressure of distributors due to drop of sales keeping the proposal of the Program, the Brazilian government decided to review the RenovaBio's mandatory targets (Fig. 3 illustrates the difference between 2nd and 3rd cycle based on 10-year period). As a result, the 2020 target reached a 50% reduction (from 29 million to 14.53 million CBIOs). Also, the Ministry of Mines and Energy (MME) and the National Energy Policy Council (CNPE) proposed a reduction in targets between 2021 and 2030, updating the growth curve for mandatory purchases according to economy and market responses as well as Covid-19 control [34].

Source: Based on [10].

In brief, reduction in mandatory targets caused by the Covid-19 pandemic, delays in the issuance of decarbonization credits (CBIOs), and an impasse with distributors are among the challenges pointed out by the sugar-energy sector in the first year of implementation of RenovaBio [10]. First of all, the RenovaBio policy was launched in 2017, but the program officially started operating in December of 2019, and it was only possible to purchase CBIOs in April 2020, as the stock market opened and negotiations began [11]. In addition, the program's targets are based on boundary conditions such as: i) Capacity of national supply of fossil fuels and biofuels; ii) Producer prices and international fuel prices; iii) Expected demand (Otto Cycle, Diesel Cycle and Aviation) in the Brazilian market; iv) Macroeconomic variables such as dollar and gross domestic production (GDP); and v) CBIO price. However, during the pandemic, these conditions were changed, and a public consultation was opened defining new targets [12,35,36].

Secondly, some issues are being discussed by the obligatory part in terms of the program's structure and achievement of targets. Also, there

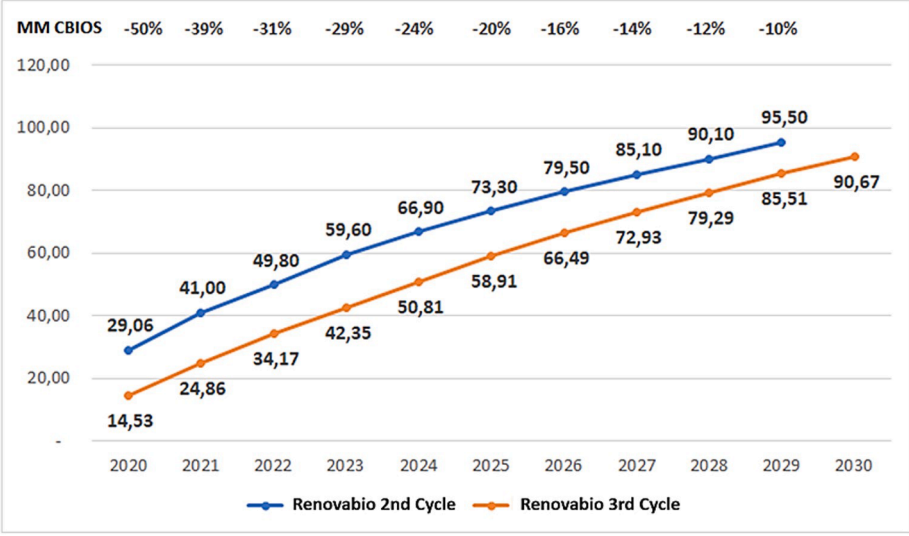


Fig. 3. Comparison of RenovaBio's targets on the 2nd and 3rd Cycle (10-year period).

are still doubts about the differentiation of taxes. It is worth mentioning that in the discussion of legal and tax aspects, the equality of taxes for both parties must prevail, in order to increase the liquidity of the paper [12,35,36].

The Covid-19 dissemination completed one year and the global economy attempted to rebound, especially by accelerating vaccine production and distribution. However, it still causes different impacts worldwide. The Oil 2021 report by the International Energy Agency (IEA) pointed out that the world oil markets are rebalancing. Nevertheless, the speed and depth of post-COVID-19 recovery may be geographically uneven [37]. In addition, the increase of ethanol prices in the beginning of 2021, which normally follows gasoline prices, also reflects the current demand and the sugarcane off-season, which already raises the price, now potentialized by a drought period in some regions of the south-central during 2020 [38]. By virtue of the drought, a drop in milling and in the production of total recoverable sugars (ATR) is expected for the 2021/22 harvest. Thus, the lower supply of agricultural feedstock, the shock in productivity and higher expenses such as transportation, will result in a 9,5% increase in the costs of sugar cane production [40].

According to Brazilian Ministry of Mines and Energy - MME [12], the first year of the program resulted in about 14 million CBIOs were retired by the obligated party, corresponding to 97.6% of the mandatory target [40] and it is projected at the end of the decade (2029) that about 590 million CBIOs should be commercialized, with an average volume of R$ 2.6 billion per year, adopting the exchange rate of 1USD = R$ 3.83. Furthermore, the estimated initial price of CBIO made by the RenovaBio Committee was close to 10 dollars, being this estimative based on experience in other countries. However, the real price is given by the market according to its variables, which the average was R$ 43,41 in 2020, with no price cap [41].

Assessment: Life Cycle analysis (LCA), land-use and production traceability

In order to issue a CBIO, biofuel producers or importers must be certified by an certifying company, which will give them an Energetic-Environmental Efficiency Score calculated using RenovaCalc, a specifically calculator that considers, beyond other aspects, the production route, the way of production, and biomass eligibility criteria. A detailed explanation of how this instrument was developed, including how the score is calculated, can be found in Scachetti et al. [21].

It is also detachable that the innovative character of RenovaBio, materialized by the RenovaCalc tool, which allows distinguishing those producers who invest in enhancing the environmental performance of their processes, is also potentially compromised by the COVID-19 crisis. The accountability of GHG emissions is based on the Life Cycle Assessment (LCA), considering the potential environmental impacts from the extraction of natural resources to produce materials to the end of the product's life (from cradle to grave). In the case of the program, the grave refers to the wheel, therefore adopting the boundary of the "well/field system to the wheel". However, due to a scenario of high uncertainties, it can compromise the carbon intensity (CI) of the biofuel.

For example, it can be mentioned the sensitivity analysis applied to sugarcane producers according to Scachetti et al. [21]. The analyses show that the CI can increase 18% in cases of low agricultural productivity (80 × 60 t/ha). The use of fertilizers, one of the biggest contributors of GHG emissions, can increase the CI by 8% depending on the nitrogen source used. This impact may be even greater in the use of RenovaCalc, if the producer opts for the standard production profile, whose certification is less costly. Furthermore, considering the national average technological level generated from sectoral database and literature data, this increase may reach more than 47% in CI. Still, the perspective of the Life Cycle is configured as a challenge for the producers regarding the traceability of the production, mainly of the soy and the corn, whose origin comes from several suppliers, when not imported. A proposal focusing on the complexity of the chain is necessary to bring transparency regarding products associated with deforestation, and confidence for the participation of soybean and corn biomass producers in the program, strengthening RenovaBio [21].

By using a calculator that considers, for instance, the biomass eligibility criteria, details of eligibility must be strictly discussed and enhanced by policymakers, producers, and suppliers. According to Cherubin et al. [26], sugarcane-derived bioenergy promotes socioeconomic development as it is a key partner to mitigate climate change and provide different ecosystem services. However, such achievements are only possible with public policies and regulations to promote rational land use and protection [42,43]. In this sense, the sucroenergy sector in Brazil doubled the area of sugarcane production. The area under sugarcane cultivation has expanded considerably, primarily driven by using ethanol as a fuel source. In addition, cultivated areas dedicated to sugarcane increased at the expense not only of pasture areas, but also of land formerly dedicated to annual and perennial food crops, such as rice, beans, corn, potatoes, cassava, and fruits [44].

Also regarding the details of eligibility, the first version of RenovaCalc considers the Agroecological Zoning of sugarcane (ZAE Cane). However, this sugarcane zoning was derogated and its requirements in RenovaCalc were removed from the calculation for the new version [45]. Regarding ZAE Cane, it was an instrument defined by the Brazilian Agricultural Policy that established criteria to order the spatial occupation by different production activities and provided guarantees that Brazilian production followed environmental requirements and rules. So, ZAE Cana takes into account criteria in order to attract investments in sugar and ethanol sectors, and its exclusion has been discussed, especially the impacts in the chain, such as the risk of biodiversity protection [46].

In this context, three main concerns rise due to this shift in agricultural land use: 1) land price increases; 2) food price increases; and 3) possible challenges to the use of sugarcane as an economical fuel source [25]. Also, RenovaBio policy fails to consider indirect-land-use GHG emissions from biofuel crops and disregards the possibility of leading deforestation by pushing other crops to new and native lands, especially in the Cerrado and the Amazon [47-50]. Lastly, discussions about measurements and traceability of agricultural feedstock, such as water, soil and fertilizer are being held.

First, RenovaBio program focuses on carbon footprint, and no water footprint was considered at the beginning. This discussion is still ongoing even though agricultural irrigation is responsible for 66% of the total water consumption [51]. Secondly, soybean and fertilizer' traceability are considered challenges because their origin comes from several suppliers or imports. According to Hilda Pereira, representing Potential Biodiesel Company, in the first year, from all certificated biodiesel plants, they could only elect 45% of their production [52]. Comparatively, this number rises to 90% in ethanol from sugarcane plants. In another case, the Bianchini Group works with soy supplied directly by around 6,500 farmers and also with grain purchased from cereal producers and cooperatives, which bring approximately 50,000 farmers, making traceability very difficult [52]. Besides, soybean has been a global commodity for several years and there is a competition when prices oscillate in the international market, opening opportunities for exportation, also potentializing a deficit in the Brazilian market and rising biodiesel prices - it is important to emphasize the need of biodiesel in the domestic market, as there are blending targets established by the National Program of Production and Use of Biodiesel [53]. Finally, as the sustainability of biofuel production becomes more relevant, the debate over fertilizers, pesticides, and other chemicals stands more powerful as they are one of the biggest contributors of GHG emissions [54]. Among sustainable initiatives to reduce these impacts are sugarcane fields using vinasse as a fertilizer [50].

In brief, public policies such as the RenovaBio program should catalyse the adoption of best practices and the production of cleaner bioenergy, improving economic, social and environmental indicators.

However, it is necessary to encourage further analyses of LCA and eligibility to check whether the policy is being implemented correctly and if it is efficient in terms of sustainability as well as compatible with other land uses [26].

Investment: Potential of bioenergy industry and regulatory aspects

RenovaBio Program presents itself as a promising, innovative policy in many aspects such as promoting biofuel expansion, boosting energy security and as a new trade market, leveraging technologies and investments. The program aims to push innovation and higher environmental responsibility, given that this would lead to the emission of additional CBIOs for commercialization [25]. For that, and in order to stimulate investments in low-carbon technologies, the Brazilian bioenergy industry is expanding first-generation sugarcane and making progress with second-generation ethanol production [55,56]. The consolidation of second-generation technologies will allow the utilization of all plant tissues, increasing ethanol productivity per hectare, besides reducing the impacts of biofuel production on food commodities [5]. However, the bottleneck now is the engineering of the mechanical process, which requires significant capital contributions - to cover the investments already made and to maintain the development of this technology [24].

Also, the generation of electricity from biomass takes place as new technologies of by-product use, such as the production of biomethane from vinasse biodigestion and biochar may represent opportunities in terms of efficiency and GHG emission reductions [26]. However, Brazil is still far from exploiting its biogas potential production [57-59]. One reason is that the market adoption grows slowly to peak in the short term. Another one is the production barrier to exhaust the amount of material available for digestion. Also, the production needs support from other sectors, such as Natural Gas, Electricity, Wastes, Water Treatment and Agriculture, in order to decrease technology prices, create favourable markets, and consequently, boost attractiveness [57].

Thus, RenovaBio creates favourable market conditions to the biogas sector, including biomethane as alternatives to provide CBIOs. It also provides the development of research about second-generation ethanol, at the same time that supports expansion and efficiency of power plants. Comparably, sugarcane is not the only option to produce bioenergy in the world, so detailed assessments are necessary to design the best option adapted to each reality. Although sugarcane-derived products provide good indicators in Brazil, it is also important to advocate for the best agricultural practices (e.g., reduced tillage, crop rotation, crop residue maintenance, and fertilization management) and improvements in by-product use e.g., vinasse, filter cake, and biochar in order to increase soil C sequestration and reduce soil GHG emissions, thus improving soil health and the sustainability of bioenergy [26]. On the other hand, policymakers are providing strategies to foment the reliability of the agribusiness, therefore investors' outlook sustain perspectives of growth, opportunities and financial returns. However, some taxes and legal aspects of RenovaBio are still being discussed, limiting the liquidity of the market and increase of different investors [60].

In this sense, Provisional Measure No. 897 of 2019, also known as "MP do Agro" (in Portuguese), was approved. It established that the revenue acquired in the negotiation of CBIOs by the producer or importer of biofuel authorized by the ANP is 15% with incidence exclusively on the source, instead of 34%. In brief, the definition of the legal nature of CBIOs is still ongoing, and it is important to measure the impacts and provide adequate treatment to all involved parties. This lack of interpretation generates uncertainties and legal insecurity for the entire commercialization chain [60].

Emission: Regulated carbon market and contributions of RenovaBio

The certification grades of biofuel facilities are inversely proportional to its carbon intensity, reflecting the individual contributions of each producer to GHG mitigation [61], and the progressively potential of CBIO emission by those facilities. For example, hydrous ethanol emits 20.8 g of CO_2 eq/MJ against 87.4 of gasoline [24], thus the difference is 55.9 g of CO_2 eq/MJ. Considering that 1 L of ethanol contains 21.35 MJ [24], each liter of ethanol mitigates approximately 1.2 kg of CO_2. As 1 CBIO corresponds to 1 t of CO_2 that is no longer emitted, 1 t of CO_2 is mitigated when 833 L of ethanol are produced [24,62,63].

As the CBIO trading market grows, questions about the use of CBIO in a regulated carbon market appear. The prospects for a carbon pricing policy have been widened with the adoption of the Paris Agreement in December 2015. While the agreement does not directly establish a global carbon pricing, discussions are being held about the provisions in Article 6 which have the potential to increase international cooperation in favour of GHG mitigation through market mechanisms. The Brazilian Nationally Determined Contribution (NDC) considers the use of such mechanisms, and among them, we have the ones considered by the new National Biofuels Policy [12,64,65].

One of the critical issues will be the definition of how mechanisms for regulated carbon markets would be applied as proposed in the Article 6 of the Paris Agreement. In this sense, RenovaBio is treated as the only national carbon pricing initiative implemented, creating expectations around its projection to the international market and development of international cooperation [66]. However, the use of CBIO in international carbon market would not be simples, as the methodology of LCA is not the commonly applied to international credits and there are different treatments for hydrous and anhydrous ethanol and biodiesel, as these two previous mentioned fuels have their own regulated mandatory blending mandates, which could hind the fungibility of credits, also generating doubts about the principle of additionality [60].

In terms of general aspects of a carbon market, it is important to reinforce the need for a national system of measurement, reporting and verification (MRV), in order to monitor, and reduce risks of carbon leakage and double emission accountability [65]. Also, it is necessary to solve governance and institutional arrangement impasses to develop the framework to promote mitigation, projects and actions, on the necessary scale in Brazil, and, in this sense, there are still institutional, taxes and legal aspects to solve in the Brazilian National Biofuel Program [67].

Finally, efforts to regulate the carbon market in Brazil must come with continued efforts to protect natural resources and mitigate deforestation. In this context, Brazil observed an increase of 9.6% in gross GHG emissions in 2019, the year in which federal climate governance was affected by the extinction of the Secretariat for Climate Change and Forests, as well as the interruption of deforestation prevention and control plans [68]. Looking at emissions at the municipal level, a research covering the period 2000 to 2018 showed that seven of Brazil's top ten GHG emitters are in the Amazon region, where forest conversion for agricultural use is behind the high rates. There is a concern about how large-scale land-use change (LUC) for the expansion of sugarcane crops potentializes side effects on environmental and socioeconomic aspects. Therefore, investments in the RenovaBio program go beyond increasing the production of biofuels, and the guidelines must include sustainable management practices as key components towards mitigation of global warming and climate change [26,69].

Recommendations

The article's discussions put light on the need to establish the legal and regulatory framework for RenovaBio, which is still under discussion. Also, the methodology of LCA and details of eligibility are relevant issues, as concerns about the shift of land use for sugarcane and traceability of agricultural feedstock, such as water, soil and fertilizer appear, also, the potential to use credits in an international carbon market. The social dimension is another point of attention, especially by the period of RenovaBio's implementation: the context of pandemic enhanced socioeconomic crises, while agribusiness faced commodities' oscillation prices as well as uncertainties about periods of isolation and fuel sales.

Possibly, it is needed a deeply debate about price cap as happens for example in Low Carbon Fuel Standard (LCFS) from California. Furthermore, RenovaBio has been fostering discussions about new technologies and development of advanced biofuels, as it creates favourable conditions for the biogas sector, including biomethane, as well as the second-generation ethanol to improve productivity per hectare and give scale to biofuels, however, it is important to attract new investments.

RenovaBio program is still crude, which gives opportunities to improvements, seeking for example: Regulate the tax structure not to create prices distortions and attract investments due to credibility of the market; establish a price-cap model or other instruments capable to support market failures; create mechanisms to give more predictability and provide transparency to CBIO offer; also, create mechanisms and policies to detail the eligibility of agricultural feedstock and land use regulations; among other aspects.

Given the relevance of Brazilian biofuels and their competitiveness, it is important to address those recommendations in order to develop a well-designed institutional framework for RenovaBio. As a result, the program can be mature enough to attend its fully goals.

Conclusion

The climate agenda and environmental pressures on the oil and gas sectors have highlighted the strategic role of the biofuels industry. Furthermore, the energy transition and security are opening a range of opportunities for this industry, especially in Brazil, a country that seeks to keep the leadership in this sector. In this context, the National Biofuels Policy was launched creating an instrument called CBIO and providing a new trade market. Such policy includes mechanisms for LCA, commercialization and predictability of the fuel market, in addition to increasing the use of biofuels, supporting national energy security and reducing GHG emissions as the policy meets Brazil's commitments under the Paris Agreement.

Even though RenovaBio potentialized the biofuel market, reducing the fossil fuel emissions in the transport sector, in addition to being the first carbon pricing initiative implemented in Brazil, there are still remaining aspects for discussion. These uncertainties highlight the need for a deep understanding of the Program, as well as the variables that directly correlate with this policy. Based on the topic-modelling methodology, our results brought relevant topics in the published articles regarding RenovaBio. In this work we focused on the discussion of the following themes: policy, assessment, investment, and emission.

Despite the macroeconomic and political situation, efforts to consolidate RenovaBio are being carried out by the government and academic institutions. Therefore, this paper brings relevant analysis for supporting discussions about biofuel policies in Brazil, especially focusing on public policies. It also encourages an open dialogue about some aspects and uncertainties, bearing in mind the need to protect natural resources and vulnerable populations, expand renewable sources to mitigate climate change impacts, and achieve the Brazilian NDC, consequently, contributing to the global mitigation target.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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